Total No. of Questions : 12]

SEAT No. :

P2144

[Total No. of Pages : 6

[5254]-540

B. E. (Mechanical SandWich) (Semester - VIII)

OPERATIONS RESEARCH

(2012 Pattern) (Elective - II)

Time : 2.½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Figures to the right indicate full marks.***
2) ***Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.***
3) ***Assume suitable data, if necessary.***

Q1) ABC company uses lathe, milling and grinding machines to manufactures parts. The given table represents the machining time required for parts. The manufacturing time available on different machines and processing on each machine part. Formulate LPP for Part I and Part II to be manufactured per week in order to maximize profit. **[8]**

Type of Machine	Machining time required for Machined parts (minutes)		Machining time required per week (minutes)
	I	II	
Lathe Machine	12	06	3000
Milling Machine	04	10	2000
Grinding Machine	02	03	900
Profit per unit	Rs 40	Rs 100	

OR

Q2) a) Explain the various environments in which decision are made. **[4]**

b) Explain basic steps involved in construction of decision tree. **[4]**

P.T.O.

Q3) Determine an initial basic feasible solution to the following transportation problem shown in table below using lowest cost entry method. **[8]**

		Warehouses						Supply
		W_1	W_2	W_3	W_4	W_5	W_6	↓
Origin	A	2	1	3	3	2	5	50
	B	3	2	2	4	3	4	40
	C	3	5	4	2	4	1	60
	D	4	2	2	1	2	2	30
Demand →		30	50	20	40	30	10	

OR

Q4) A company has one surplus car in each of the district's 1, 2, 3, 4 and 5 and one deficit car in each of districts I, II, III, IV, V and VI. The distance between districts in kilometers is given in table below. Determine the assignment of cars from districts in surplus to districts in deficit so that total distance covered by car is minimum. **[8]**

	I	II	III	IV	V	VI
1	12	10	15	22	18	8
2	10	18	25	15	16	12
3	11	10	3	8	5	9
4	6	14	10	13	13	12
5	8	12	11	7	13	10

Q5) Using method of sub game, solve the following game **[6]**

		Player 2		
		1	2	3
Player 1	1	1	3	11
	2	8	5	2

OR

Q6) XYZ production company have following data,

Sell price per unit = Rs 14

Total units sold = 50,000

Fixed cost = Rs12,000

Find: i) P/V ratio

ii) B.E.P. in units

iii) B.E.P. in sales

iv) Margin of safety

v) Total profit

[6]

Q7) a) A chemical company requires 3600 kg of raw material for manufacturing a particular item/year. It has been estimated that cost of placing an order is Rs 36 and cost of carrying an inventory is 25% of investment in inventories. The price is Rs 10/kg.. **[8]**

Find i) optimal lot size ii) optimal order cycle time

iii) minimum yearly total cost

b) An Automotive company is thinking of replacing a particular machine whose cost price is Rs 12,200. The scrap value of the machine is Rs 200/-. The maintenance costs are found to be as follows : **[8]**

Year	1	2	3	4	5	6	7
Maintenance cost in Rs	220	500	800	1200	1800	2500	3200

Determine when the company should get the machine replaced?

OR

Q8) a) A manufacturing company purchases 9000 parts of a machine for its annual requirements, ordering one month's requirement at a time Each part costs Rs 20 The ordering cost per order is Rs15 and carrying charges are 15% of average inventory/year Suggest a economical purchasing policy for company. What advice would you offer and how much would it save the company per year? **[8]**

b) Machine X costs Rs 45,000/- and the operating costs are estimated at Rs 1000/-for the first year, increasing by Rs 10,000/- per year in the second and subsequent years. Machine Y costs Rs 50000/and operating costs are Rs 2000/- for the first year, increasing by Rs 4000/- in the second and subsequent years. If we now have a machine of type X, should we replace it by Y. If so when? Assume both machines have no resale value and future costs are not discounted. **[8]**

Q9) a) A machinist finds that the time spent on his jobs have an exponential distribution with mean of 30 minutes. If he repairs sets in the order in which they come in, and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day, what is machinist's expected idle time each day? How many jobs are ahead of the average set just brought in? **[8]**

b) Five jobs A, B, C, D and E are to be made on the three groups of machines m_1, m_2 and m_3 in that order, the time required for each job is given in table below Sequence the job on machines for minimum elapsed time starting from first job on m_1 to the finishing of last job on m_3, also find idle time for machine during elapsed time. **[8]**

Job	A	B	C	D	E
Time for m_1 (min.)	20	27	31	15	19
Time for m_2 (min.)	7	9	6	12	14
Time for m_3 (min.)	27	31	16	14	16

OR

Q10) a) In a bank one cashier is there to serve the customers. And the customers pick-up their needs by themselves. The arrival rate is 9 customers for every 5 minutes and the cashier can serve 10 customers in 5 minutes. Assuming Poisson arrival rate and exponential distribution for service rate. Find : **[8]**

i) Average number of customers in the system.

ii) Average number of customers in the queue or average queue length.

iii) Average time a customer spends in the system.

iv) Average time a customer waits before being served

b) There are five jobs each of which must go through two machines X and Y in the order of XY. Processing times are given in table below. Determine a sequence for five jobs that will minimize the elapsed time and also calculate the total idle time. **[8]**

Job	1	2	3	4	5
Time for X (min.)	5	1	9	3	10
Time for Y (min.)	2	6	7	8	4

Q11) a) A project consists of six activities. Draw the network diagram and calculate EST, LST, EFT, LFT and floats. Determine the critical path and find total project duration. **[8]**

Activity	Immediate Predecessor	Activity Duration
A	-	4
B	A	6
C	B	5
D	A	4
E	D	3
F	E, C	3

b) An electronic company sells some good items, the post data of demand per week in 100 kg with frequency is given below. **[8]**

Demand	0	5	10	15	20	25
Frequency	2	11	8	21	5	3

Using the following sequence of random nos. 35, 52, 19, 13, 23,93, 34, 57, 35, 83. Generate demand for next 10 weeks and also find average demand per week.

OR

Q12) a) A small maintenance project consist of following 12 jobs. Draw the network of the project. Summarize CPM calculations in tabular form. Calculating the three types of floats for jobs and hence determine critical path. **[12]**

Job	1-2	2-3	2-4	3-4	3-5	4-6	3-8	6-7	7-9	6-10	8-9	9-10
Duration	2	7	3	3	5	3	5	8	4	4	1	7

b) A T.V. company produces two components, A and B which must be processed through assembly and finishing department. Assembly has 90 hours available, finishing can handle upto 72 hours of work. Manufacturing one A component needs 6 hours in assembly and 3 hours in finishing. Each B components needs 3 hours in assembly and 6 hrs in finishing, If profit is Rs120 per A and Rs 90 per B, calculate the best combinations of A and B to realize profit of 'Rs 2100 **[4]**